

Viral Structure & Pathogenesis - Factoid Sheet

Week 8 • Core Lecture 3 • current Thai deck controls

Framework: structure → receptor/tropism → replication strategy → spread → cell/immune injury → diagnostic morphology.

Envelope vs Naked

Enveloped	Naked
Host-derived lipid membrane + viral glycoprotein spikes	Capsid is outermost layer
Fragile: acid/detergent/drying/heat sensitive	Stable: relatively resistant
Usually body-fluid transmission; poor GI survival	Readily fecal-oral; survives GI acid/bile
Budding or lysis	Lysis

Life Cycle + Receptor Tropism

Attachment → entry → uncoating → replication/transcription → protein synthesis → assembly/maturation → release.

Virus	Deck receptor
EBV	CR2 on B cells
HIV	CD4 + CCR5/CXCR4
Parvovirus B19	Globoside / P antigen on proerythroblasts
SARS-CoV-2	ACE2

DNA vs RNA Rules

DNA Usually nucleus. Pox is classic cytoplasmic exception. HBV is special because it uses an RNA intermediate and reverse transcriptase.	RNA Usually cytoplasm. Influenza and retroviruses are nuclear exceptions. +RNA can act as mRNA; -RNA requires viral RdRp.
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Influenza: 8 segments. Drift = slow change/yearly outbreaks. Shift = reassortment/pandemic potential.

Medically Important Virus Anchors

Virus/family	High-yield lecture anchor
Herpesviridae	HSV, CMV, EBV; latency
HBV	Reverse-transcribing DNA virus; HCC association
HPV	6/11 warts; 16/18 cervical carcinoma
Parvovirus B19	Fifth disease; fetal anemia/hydrops
Influenza A	8 segments; drift/shift
Retrovirus / HIV	RNA → DNA → host-genome integration

Pathogenesis + Histopathology

Transmission / Progression

- Portal: broken skin or mucosa; inhalation examples include influenza and SARS-CoV-2.
- Bloodstream spread = **viremia**.
- Acquisition → incubation/amplification → prodrome → tissue-specific disease → immune response → shedding → resolution/persistence.

How Viruses Injure Cells

Pattern	Examples in deck
Syncytium	HSV, VZV, HIV, RSV, SARS-CoV-2
Chromatin margination	Parvovirus B19, HSV, adenovirus
Inclusion bodies	Location/stain can identify virus
Koilocytosis	HPV
Immune-mediated injury	Interferon/cytokines, T-cell injury, immune complexes, immunosuppression

Must-Know Morphology

Virus	Hallmark
Rabies	Negri body - eosinophilic intracytoplasmic neuronal inclusion
CMV	Owl's-eye basophilic intranuclear inclusion
HSV/VZV	Cowdry eosinophilic intranuclear inclusions + chromatin margination
Parvovirus B19	Chromatin margination; fifth disease/hydrops context
HPV	Koilocytosis
RSV	Syncytium formation in pediatric pneumonia

EBV objective gap: current deck has no dedicated EBV histology image. MCG 2029 handout adds atypical lymphocytes (Downey cells) as its recognition clue.

General Diagnosis - Handout Enrichment

Test	Use in handout
PCR	Detect viral nucleic acid; highly sensitive/specific
Antigen test	Detect viral proteins; rapid but often less sensitive
Serology	IgM recent/acute; IgG prior infection/vaccination

SARS-CoV-2 Lecture Model

ACE2 binding / loss of ACE2 activity → ↑ Ang II → vasoconstriction + coagulopathy + alveolar injury → ARDS / myocardial injury / renal failure.	
Agent	Deck mechanism
Remdesivir	RdRp inhibitor
Nirmatrelvir/ritonavir	Protease inhibitor + CYP3A4 inhibition
Molnupiravir	Nucleoside analog; deck describes RdRp inhibition
Vaccines	mRNA; protein subunit

Final buzzwords: Negri = rabies • owl's eye = CMV • Cowdry = HSV/VZV • koilocyte = HPV • syncytium = RSV • P antigen = B19 • ACE2 = SARS-CoV-2.

Primary: current Thai 49-slide deck. Official Week 8 LO sheet supplies 4 ScholarRx bricks. MCG 2029 AI Notes handout is secondary enrichment only.